PHYSICAL CHEMISTRY CUMULATIVE EXAM

January 21, 2000

The molecular vibrational partition function can be written as:

$$q_v = [1-\exp(-h\nu/kT)]^{-1}$$

where ν is the harmonic oscillator frequency, $k = 1.381 \times 10^{-23}$ J/K, T is the temperature, and $h = 6.626 \times 10^{-34}$ Js. The harmonic oscillator frequency for iodine, I_2, is 214.6 cm^{-1}; for hydrogen, H_2, it is 4160 cm^{-1}. The speed of light is $c = 2.998 \times 10^8$ m/s.

1.) In general, how is a molecular partition function defined? (That is, specify the mathematical operations that will lead to the formula given above.) (15%)

2.) Some insight into the significance of a partition function can be obtained by considering how it depends on the temperature, T. In general, what happens to a molecular partition function as $T \rightarrow 0$, and $T \rightarrow \infty$? (15%)

3.) The molecular partition function allows one to calculate the fractions of molecules in state i, $p_i = n_i/N$, where n_i is the number of molecules in state i, and N is the total number of molecules.

For I_2 use the information given above to calculate how many of its vibrational energy levels have to be populated to contain more than 98% of a gaseous sample at room T, 25°C. (40%)

4.) Compare the result for I_2 calculated above with the same calculation for H_2. (30%)